

Consensus Dietary Guidelines for Healthy Living and Prevention of Obesity, the Metabolic Syndrome, Diabetes, and Related Disorders in Asian Indians

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Abstract

India is undergoing rapid nutritional transition, resulting in excess consumption of calories, saturated fats, *trans* fatty acids, simple sugars, salt and low intake of fiber. Such dietary transition and a sedentary lifestyle have led to an increase in obesity and diet-related non-communicable diseases (type 2 diabetes mellitus [T2DM], cardiovascular disease [CVD], etc.) predominantly in urban, but also in rural areas. In comparison with the previous guidelines, these consensus dietary guidelines include reduction in the intake of carbohydrates, preferential intake of complex carbohydrates and low glycemic index foods, higher intake of fiber, lower intake of saturated fats, optimal ratio of essential fatty acids, reduction in *trans* fatty acids, slightly higher protein intake, lower intake of salt, and restricted intake of sugar. While these guidelines are applicable to Asian Indians in any geographical setting, they are particularly applicable to those residing in urban and in semi-urban areas. Proper application of these guidelines will help curb the rising “epidemics” of obesity, the metabolic syndrome, hypertension, T2DM, and CVD in Asian Indians.

Introduction

ASIAN INDIANS (people of Indian origin living in India or living in other countries) have become more affluent, urbanized, and mechanized during the previous decade. A hectic lifestyle and the easy availability of convenience foods have led to irregular meals and frequent snacking on energy-dense fast foods (“fast foods” refer to energy-dense foods prepared and sold commercially by roadside vendors and food outlets, prepared either by deep frying or with preheated or precooked ingredients; these foods typically have low nutritional value and preparation time), including ready-to-use gravies and soups, packaged salty snacks, ready-made cookies, and commercial fast-foods rather than traditional home-cooked food.¹ Furthermore, consumption of animal foods, sweetened carbonated drinks, sugar, and sweeteners

(Table 1) has also increased.³ In addition, traditional Indian energy-dense foods continue to be consumed. Overall, this nutritional transition, particularly noticeable in children, has resulted in high consumption of calories, saturated fats (Table 2), *trans* fatty acids (TFAs), simple sugars, and salt, along with low intake of fiber, monounsaturated fatty acids (MUFAs), and *n*-3 polyunsaturated fatty acids (PUFAs).¹ This nutrition transition has the potential to cause obesity and other diet-related non-communicable diseases (DR-NCDs) such as type 2 diabetes mellitus (T2DM), hypertension, and cardiovascular disease (CVD),³ predominantly in urban areas but also in semi-urban and rural areas.^{6,7}

Despite possible influences of genetic and perinatal factors, imbalanced diets and physical inactivity (Table 3) are likely to have greater and overriding influence on the increasing prevalence of obesity in India.^{3,12} Persistent obesity

¹Affiliations and group members are given in Appendix 1.

TABLE 1. SECULAR TRENDS OF NUTRIENT CONSUMPTION IN INDIA

Product	Average values ^a		
	1979–1981	1989–1991	1999–2001
Total animal product	120	163	196
Animal fat	23	28	47
Eggs	3	5	6
Seafood	5	7	8
Meat	16	20	22
Milk-excluding butter	71	102	111
Total vegetable product	1,963	2,202	2,296
Alcoholic beverages	5	8	11
Cereals	1368	1508	1470
Fruits	31	34	51
Oil crops	25	37	43
Pulses	120	133	109
Rice (milled)	670	779	751
Starchy roots	41	40	49
Sugar and sweeteners	193	221	247
Sugar crops	8	9	11
Vegetable oils	127	158	239
Vegetables	32	35	45
Wheat	390	461	493
Grand total	2,083	2,365	2,492

Data are obtained from the Food and Agricultural Organization Database (FAOSTAT).²

^aAll variables expressed in the unit calories/capita/day.

dysregulates metabolic processes, including action of insulin on glucose–lipids–free fatty acid metabolism, causing clustering of dysglycemia, dyslipidemia, hypertension, and the procoagulant state, known as the metabolic syndrome. Obesity and the metabolic syndrome are immediate precursors of T2DM and CVD.¹³ The prevalence of insulin resistance and the metabolic syndrome is rapidly increasing in urban areas in India. The severity of insulin resistance and related cardiovascular risk factors is higher in Asian Indians than white Caucasians.³ Furthermore, T2DM and CVD occur a decade earlier in Asian Indians than in white Caucasians and lead to more complications.¹⁴ The magnitude of T2DM, CVD, and other metabolic disorders, however, varies according to the affected populations (rural or urban), socioeconomic strata, and region of residence in India.¹⁵ The increasing prevalence of DR-NCDs in the Asian Indian population is of great concern, necessitating preventive steps.

Why Revisions in Dietary Guidelines Are Needed

The following factors have necessitated revision of existing dietary guidelines. Moreover, the present guidelines specifically focus on healthy living and on prevention of DR-NCDs in Asian Indians:

1. Recent dietary transition in India as discussed above^{3,16}
2. Rapid increase in DR-NCDs in India¹
3. New research data on macronutrients and micronutrients in Indian diets
4. Need to acquire user-friendly dietary guidelines that could be understood and used not only by nutritionists and general physicians but also by the general population

TABLE 2. CONSUMPTION OF FATS AND FATTY ACIDS IN URBAN ADOLESCENTS AND ADULTS IN INDIA

Age group	Total fat (% energy) (RDA 15–30%)						SFA (% energy) (RDA <10%)						PUFA (% energy) (RDA <8%)						ω-3 PUFA (% energy) (RDA not <1%)						ω-6 PUFA (% energy) (RDA 3–7%)						MUFA (% energy) (RDA <10%)						TFA (% energy) (RDA <1%)					
	Male			Female			Male			Female			Male			Female			Male			Female			Male			Female			Male			Female								
	M	SD		M	SD		M	SD		M	SD		M	SD		M	SD		M	SD		M	SD		M	SD		M	SD		M	SD										
Adolescents 13–18 years (n = 797) ^a	32.4	7		35.6	6.3	11.6	4.0	10.7	3.8	7.9	3.2	10.2	3.7	0.9	0.6	1.2	0.8	7.2	3.3	9.2	3.7	9.3	2.5	10.1	2.8	1.1	(0–10.7)	1.1	(0–10.2)													
Adults 19–49 years (n = 325) ^a	30.8	8.4		34.1	6.7	8.9	3.7	9.3	3.1	9.5	3.6	10.1	3.6	1.4	0.9	1.9	1.0	8.2	3.8	8.3	3.8	9.6	3.1	11.6	3.4	0.4	(0–4.1)	0.6	(0–9)													
>50 years (n = 124) ^a	31.2	6.8		33.5	6.2	9.2	3.2	9.7	3.7	9.7	3.4	9.9	3.8	1.0	0.7	1.5	1.2	8.7	3.7	8.4	4.0	9.2	2.8	10.5	3.4	0.3	(0–6.2)	0.4	(0–6.8)													
18–69 years (n = 227) ^b	24.7			(12–78.9)	28.7	6.6	(0.9–21.5)	6.5	(1.1–22.1)	—	—	—	—	1.3	0.6	1.7	0.8	3.1	1.1	3.8	3.1	4.7	(1.4–16.1)	5.7	(1.2–17)	1.0	5.2	0.8	1.7													

Table adapted from Misra et al.¹ Mean (M) and SD values are given, and numbers in parentheses denote the range.

^aRepresentative sample from general population in New Delhi, India (authors' unpublished results).

^bPeople belonging to low socioeconomic stratum living in urban slums, New Delhi.⁴ PUFA, polyunsaturated fatty acids; RDA, recommended dietary allowance for Asian Indians;⁵ SFA, saturated fatty acids; TFA, trans fatty acids.

TABLE 3. DIFFERENCES IN PHYSICAL ACTIVITY AMONG ASIAN INDIANS VERSUS WHITES/EUROPEANS

<i>Study</i>	<i>Ethnic group</i>	<i>Physical activity criteria/parameter</i>	<i>Asian Indians</i>	<i>Whites/Europeans</i>
Mohanty et al. ⁸	Asian Indians (<i>n</i> = 555), non-Hispanic whites (<i>n</i> = 87,846)	Reporting vigorous activity	33%	40.7%
Shaukat et al. ⁹	Asian Indians (<i>n</i> = 89), Caucasians (<i>n</i> = 82)	Physical activity index	8.5 ^a	13.7
Dhawan et al. ¹⁰	Asian Indians (<i>n</i> = 80), Caucasians (<i>n</i> = 82)	Exercise for at least 20 min once a week	17%	34%
Petersen et al. ¹¹	Asian Indians (<i>n</i> = 49), Caucasian children (<i>n</i> = 292)	Physical activity index	2.1 ^a	2.3

^aUnit not given as this is a ratio.

- Need to evolve food-based dietary guidelines to translate nutrient recommendations into foods for easy interpretation.

The Consensus Development Process and Preparation of This Article

The objectives of this consensus process are to critically analyze current nutrition transition and based on this analysis, revise the previous dietary guidelines for India (prepared in 1998 by the National Institute of Nutrition), specifically for healthy living for Asian Indians and prevention of DR-NCDs, and to make dietary guidelines user-friendly.

Experts (see Appendix 2) from across the country and belonging to the various related fields, including nutrition, internal medicine, diabetes, metabolism, endocrinology, cardiology, exercise physiology, and sports medicine and representing reputed medical institutions, hospitals, government-funded research institutions, and policy-making bodies participated in a "Consensus Summit" to develop Asian Indian-specific dietary guidelines. All the research groups in India doing original research on nutrition were contacted and invited. The Steering Committee (see Appendix 2) prepared a draft document well in advance of the summit, which was communicated to all prospective participants for feedback and comments. After the valid suggestions were incorporated, the revised consensus document was circulated among all the experts for a second review before the consensus meeting (April 5, 2009, in New Delhi). The experts appraised the rationale, background, and proposed changes in the form of five lectures, followed by two panel discussions steered by four or five reputed experts, during the summit. Discussions were held based on the following questions/issues, taking into account the research data in Asian Indians published from India and elsewhere:

- What should be the recommendation(s) for the following?
 - Carbohydrates and fiber
 - Proteins
 - Fats and TFAs
 - Quality and quantity of cooking oils
 - Water intake
 - Alcohol intake
 - Salt and sugar intake
- Choice of foods while eating out
- Meal timings and gaps between meals, and other dietary habits
- Cooking methods

The summary points were presented to the whole group and unanimously adopted as consensus guidelines. A writing group led by experts who have done research on specific nutrients worked on the article (see Appendix 2). The draft of the manuscript was again circulated by e-mail to all (even those who did not participate [see Appendix 2]) for any editing before the publication.

The literature search was done using the key words "Diet and Asian Indians or dietary recommendations for Asian Indians, or trends in dietary intake in Asian Indians" from the medical search engine PubMed (National Library of Medicine, Bethesda, MD) from 1966 to October 2009. Manual searches for other important references and nutritional databases were also conducted. Many of the food articles have been named both in English and in Hindi (given in italics).

The Secular Trends in Intake of Macronutrients in India

Data regarding time trends in food and nutrient intakes are available from the National Nutrition Monitoring Bureau, which was established in each of 10 major states in India in 1972. Repeat National Nutrition Monitoring Bureau surveys have shown a decrease in the average intake of all the macronutrients between 1975–1979 and 1996–1997 among rural adults. The intake of protein was 62 g/consumption unit/day in 1975–1979 and declined to 54 g in 1996–1997 among rural adults ≥ 18 years age; energy also declined from 2,350 Kcal in 1975–1979 to 2,110 Kcal in 1996–1997.¹⁷ A similar trend was observed among tribal population in various parts of India: protein, 55 g/day in 1985–87 and 54 g/day in 1998–1999; energy, 2,213 Kcal/day in 1985–1987 and 2,239 Kcal/day in 1998–1999.¹⁸ While carbohydrates remain the major source of energy in Indian diets, the percentage of total energy intake derived from carbohydrates has declined (1975–1979, 80.3%; 2001, 75.5%), with an increase in the percentage of energy coming from dietary fats (1975–1979, 8.9%; 2001, 13.9%). However, the proportion of dietary energy from fat still remains less than 15%, which is lower than the recommended dietary allowance (RDA) of 15–30%.¹⁹ Consumption of oils, fats, and animal products has increased in almost all the states.¹⁷ Energy intake is lower in urban areas, in spite of higher intake of fats and oils, because of lower cereal consumption compared with rural areas.²⁰ Several reasons attributable for these dichotomous observations of decreased energy intake with rising prevalence of obesity, could be: under-reporting of dietary consumption data,²⁰ higher energy intake in comparison with energy expenditure,²¹ and increasingly sedentary lifestyle.¹⁷

Energy Recommendation

The recommended energy should be adequate to maintain ideal weight and health in adults. If body weight and physical activity (Tables 4 and 5) of an individual are known, it is easy to calculate the extra needs of energy for a particular situation. In the case of energy, the RDA represents only the average daily requirement corresponding to daily average expenditure of an individual. The energy requirements are suggested based on type of activity profile (sedentary, moderate, and heavy), age, gender, and physiological status of an individual.²⁴ Energy requirement for any individual is calculated by multiplying the activity factor by ideal body weight of that individual (Tables 4 and 5). For example, an Asian Indian man with medium body frame, 165 cm tall, should ideally weigh 62 kg and would require 1,850 Kcal to maintain a healthy weight if that person is sedentary. Ideal body weight should be aimed to maintain a body mass index between 18 and 22.9 kg/m².²⁵

Carbohydrates and Fiber

Carbohydrates are divided into simple and complex carbohydrates. Simple carbohydrates (like sugar, refined flour, candies, toffees, etc.) should be avoided as they break down much faster and cause insulin levels to spike quickly. Complex carbohydrates (whole cereals, unpolished rice, barley [*jaun*], buckwheat [*kuttu*], oats [*jai*], millets, etc.) are subdivided as starches and fiber, are best consumed in the unprocessed form, and should be the principal source of energy. Diet high in natural fiber ameliorates the slow release of glucose in the blood from the intestine and helps in regulating blood glucose levels. The RDA for carbohydrates should be 50–60% of total energy, which would ensure an appropriate macronutrient balance. While deciding for carbohydrates, the glycemic index (GI) of foods should also be considered. GI is a measure of the effects of carbohydrates on blood sugar levels. Carbohydrates that break down quickly during digestion and release glucose rapidly into the bloodstream have a high GI, whereas carbohydrates that break down more slowly, releasing glucose more gradually into the bloodstream, have a low GI. Emerging research, globally and from India, has shown the relevance of GI in the Indian context.^{26–29} The amount of carbohydrate consumed also affects blood glucose levels and insulin responses. The glycemic load of a food is calculated by multiplying the GI by the amount of carbohydrate (in g) provided by a food and dividing the total by 100. Although fructose has a low GI, recent research indicates that eating too much fructose may increase the rate of liponeogenesis and lipid storage in the liver. Hence intake of processed fructose or high fructose corn syrup-containing foods and drinks should be minimized.

The Dietary Approaches to Stop Hypertension (DASH) trial demonstrated that a carbohydrate-rich diet consisting of whole grains, fruits, vegetables, and low-fat dairy products, low in saturated fat, total fat, and cholesterol, substantially lowered blood pressure and low-density lipoprotein cholesterol level.³⁰ At least four or five servings of fruits and vegetables (rich sources of fiber) in the diet also ensure adequate intake of micronutrients, including antioxidants, fiber, and phytonutrients. The World Health Organization recommends an intake of a minimum of 400 g of fruits and vegetables per day for the prevention of chronic diseases.³¹

Key principles

1. Eat complex carbohydrates with low GI instead of simple sugars or high GI carbohydrates.
2. Eat high fiber foods.
3. Quantity and quality of carbohydrates are equally important in the daily diet.

Recommendations

1. The daily carbohydrate intake should be approximately 50–60% of the total calorie intake. For example, in 1,800 and 2,000 calorie diets, the carbohydrate intake for a sedentary to moderately active individual should be 225–270 g/day and 250–300 g/day, respectively.
2. The primary source of complex carbohydrates in the diet should be cereals (whole wheat, brown rice, etc.), millets (pearl millet [*bajra*], finger millet [*ragi*], great millet [*jowar*]), pulses (red gram [*tur dal*], green gram [*sabut moong*], etc.), and legumes (soya, horse gram [*kulthi*]). Complex carbohydrates should be preferred over refined carbohydrates and its products (e.g., whole grain bread over white [*maida*] bread).
3. Low GI foods (e.g., oats [*jai*], unpolished rice, parboiled rice, whole pulses, beans [*fali*], and legumes (*sabut anaz*) and some whole fruits (like guava, apple, etc.)) should be preferred. High GI foods (refined flour, root vegetables such as yam [*sooran/shakarkand*], potato, tapioca [a type of *shakarkand*], colocasia [*arbi*], etc.) should be consumed in moderation.
4. The total dietary fiber in daily diet should be 25–40 g/day (e.g., 100 g of apple [1 small apple] gives 1.0 g of fiber; 100 g of whole wheat flour gives 1.9 g of fiber). Whole grains, cereals, pulses, vegetables, and fruits contain high dietary fiber.
5. A minimum of four or five servings per day of fruits and vegetables is recommended (i.e., approximately 400–500 g/day including three vegetable and two fruit portions [e.g., 100 g of (one *katori*) raw vegetables (e.g.,

TABLE 4. CALCULATION OF IDEAL BODY WEIGHT

Build	Women	Men
Medium	100 lbs (45.5 kg) for the first 5 feet (152 cm) height, plus 5 lb (2.3 kg) for each additional inch	106 lbs (48 kg) for the first 5 feet (152 cm) of height, plus 6+lbs (2.7 kg) for each additional inch
Small	Subtract 10%	Subtract 10%
Large	Add 10%	Add 10%

Adapted from the 1977 publication by the American Diabetes Association and the American Dietetics Association.²²

TABLE 5. CALCULATION OF ENERGY REQUIREMENT

Activity level	Energy requirement (Kcal/kg of IBW/day)		
	Obese	Normal	Underweight
Sedentary	20–25	30	35
Moderate	30	35	40
Heavy	35	40	45–50

According to Williams.²³
IBW, ideal body weight.

cauliflower, brinjal, etc.) = 20–30 Kcal; 100 g of fruit e.g., one apple = 59 Kcal]). Fruits should be eaten whole, preferably with the skin, whenever feasible instead of fruit juices.

- Simple sugars like crystalline sugar, sugarcane juice, sweetened carbonated beverages, fruit juices, and sugar syrups should be avoided.

Fats

A small amount of fat is present in almost every food item (invisible fat). The fat in processed and ready-to-eat foods (hidden fat) and visible fat (vegetable oil, *ghee*, and *vanaspati*), used for cooking together contribute to the total fat intake of an individual. Dietary fats (lipids) are important components of human diet, providing energy and essential fatty acids (linoleic acid [LA] and α -linolenic acid [ALA]) and serving as a source of fat-soluble vitamins (e.g., vitamins A, D, E, and K). Fats improve texture and palatability of foods and have an important role in inducing satiety. The nutritional and health effects of dietary fats are determined by the nature of their constituent fatty acids (>95%) and the composition of minor components (<5%), collectively designated as “nonglyceride components.”

Fatty acids are categorized as follows: saturated fatty acids (SFAs), MUFAs, and PUFAs. All fatty acids generate energy and when taken in excess are stored in adipose tissue. The nutritionally significant and naturally occurring MUFAs and PUFAs have a *cis* configuration. TFAs are mainly generated by industrial partial hydrogenation of vegetable oils. In addition, small amount of TFAs are present in ruminant fats by bacterial hydrogenation of PUFAs in the rumen of ruminants (e.g., cattle, buffalo, camels, goats, etc.). The industrial hydrogenation process results in formation of predominantly elaidic acid (20–50%) as the *trans* isomer.

The pathological processes involved in the causation and complications of coronary heart disease comprise disturbances in lipoprotein metabolism (high levels of low-density lipoprotein, low levels of high-density lipoprotein, and high levels of serum triglycerides), endothelial dysfunction, increase in pro-inflammatory cytokines, prothrombotic shift in arterial homeostasis [high levels of lipoprotein (a) and thromboxane/prostacyclin ratio], and insulin resistance.^{31,32} SFAs raise total and low-density lipoprotein cholesterol levels, reduce insulin sensitivity, and enhance thrombogenicity, thus contributing to an increase in coronary heart disease risk. TFAs increase low-density lipoprotein cholesterol (an effect similar to SFA), decrease high-density lipoprotein cholesterol (an effect different from SFA), and

increase the total cholesterol/high-density lipoprotein cholesterol ratio, a powerful predictor of increased risk of coronary heart disease. The consumption of excess TFAs increases coronary heart disease risk to a greater extent than diets with excess SFAs.³¹ The intake of ruminant TFAs is low compared with those obtained from partial hydrogenation of vegetable oils, and hence the focus should be on reducing TFAs from partial hydrogenation of vegetable oils.^{33,34} PUFAs enhance peripheral glucose utilization, improve insulin action, and reduce adiposity. Recent evidence suggests that diets providing high intakes of both PUFAs (LA and ALA), balanced LA/ALA ratio, and long-chain *n*-3 PUFAs from fish/fish oils prevent DR-NCDs.^{35–37} Hence, optimal intakes of these fatty acids may be compromised with low fat diets. Plant sterols and other unique components (oryzanols and sesame lignans) in the nonglyceride component of fats in foods and vegetable oils also contribute to lowering of low-density lipoprotein cholesterol.³⁸

Key principles

- The lower limit of fat should be adequate for the energy needs (15% of total energy), should prevent essential fatty acid deficiency (LA, 3% of total energy; ALA, 0.5% of total energy), and should facilitate optimal absorption of fat-soluble vitamins.³⁹
- High levels of SFAs and TFAs promote dyslipidemia and atherosclerosis.

Recommendations

For optimal health across the life course the following recommendations, along with food-based guidelines, are suggested:^{31,36}

- Fats should provide not more than 30% of total energy/day and SFAs should provide no more than 10% of total energy/day. For individuals having low-density lipoprotein cholesterol of ≥ 100 mg/dL, SFAs should be <7% of total energy/day.
- Essential PUFAs (LA) should provide 5–8% of total energy/day.
- ALA should be 1–2% of total energy/day.
- The optimal ratio of LA/ALA should be 5–10.
- Long-chain *n*-3 PUFAs should be obtained from fish, walnuts, flaxseeds, canola oil, etc.
- Cis* MUFAs should provide 10–15% of total energy/day.
- TFAs should be <1% of total energy/day.
- Cholesterol intake should be limited to 200–300 mg/day.

Food-based guidelines to ensure optimal fat quality in Asian Indian diets

- Complete dependence on just one vegetable oil does not ensure optimal intake of various fatty acids (Table 6), use of two or more vegetable oils is recommended.
- The recommendation for oils are as follows:⁴¹
 - Preferred vegetable oil(s) along with ALA-containing oil(s) or vegetable oil containing high LA levels along with oil(s) containing moderate or low LA levels are listed in Table 7. However, the latter

TABLE 6. APPROXIMATE FATTY ACID COMPOSITION OF COMMONLY AVAILABLE FATS AND OILS

Fats/oils	(% of total fatty acids)			
	SFAs	MUFAs ^a	LA	ALA
High medium-chain SFAs				
Coconut	92	6	2	—
Palm kernel	83	15	2	—
Butter/ <i>ghee</i> *	68	29 ^b	2	1
High MUFAs				
Olive	16	71	12	1
High SFAs and MUFAs				
Palmolein	39	46	11	<0.5
High MUFAs and moderate LA				
Groundnut	19	41	32	<0.5
Rice bran	17	43	38	1
Sesame	16	41	42	<0.5
High LA				
Cottonseed	24	29	48	1
Corn	12	35	50	1
Safflower	9	13	75	—
Sunflower	12	22	62	—
LA (<i>n</i> -6) and ALA				
Soybean	14	24	53	7
Canola	6	60 ^c	22	10
Mustard/rapeseed	4	65 ^d	15	14
Flaxseed	10	21	16	53
High TFAs				
<i>Vanaspati</i> **	46	49 ^e	4	—

Data are from the 1996 Codex Alimentarius commission report.⁴⁰

^aMainly oleic.

The following superscripts indicate the percentage of TFA: ^b5%, ^c7% (range, 5–38% for Indian data compiled between 2000 and 2009).

The following superscripts indicate the percentage of erucic acid: ^c~2%, ^d~50%.

*Clarified butter.

**Hydrogenated fat.

ALA, α -linolenic acid; LA, linoleic acid; MUFAs, monounsaturated fatty acids; PUFAs, polyunsaturated fatty acids; SFAs, saturated fatty acids; TFAs, *trans* fatty acids.

combination would ensure moderation in LA intake only and is recommended when other dietary components provide high ALA levels or fish is consumed. Improvement of *n*-3 PUFA nutritional status in Indian adults has been shown with two of these oil combinations (groundnut oil/sunflower oil and canola).⁴¹

- b. Consumption of butter and *ghee* (clarified butter) should be kept to a minimum.
 - c. Use of partial hydrogenation of vegetable oils (*vanaspati*) as the cooking medium should be strictly avoided.
 - d. Coconut oil, palm kernel oil, palm oil, and palmolein or their solid fractions should be substituted for partial hydrogenation of vegetable oils in foods that require solid fats (bakery fats, shortening, etc.). These oils are high in SFAs but are TFA free.
3. To ensure correct balance of fatty acids from dietary components other than visible fat, the following dietary guidelines are recommended:^{41–43}
 - a. Regular consumption of foods with high ALA content (wheat, pearl millet, pulses, green leafy vegetables, fenugreek, flaxseed, and mustard seeds) (Table 8).
 - b. Partial replacement of visible fat and invisible fats from animal foods with whole nuts such as pistachios and almonds.
 - c. Moderation in the use of animal foods containing high levels of fat, SFAs, and cholesterol.
 - d. For nonvegetarians, consumption of 100–200 g of fish (four to six pieces)/week.
 - e. Minimizing consumption of premixed, ready-to-eat, fast foods, bakery foods, and processed foods prepared in partial hydrogenation of vegetable oils like savory (*namkeen*).
 - f. Choose low fat dairy foods such as double-toned milk (fats <1.5%) or curd prepared from such milk. The preference for low fat dairy foods would also reduce ruminant TFAs.

Proteins

The protein requirement as suggested by the 2007 World Health Organization/Food and Agriculture Organization/United Nations University Expert Consultation is 0.66 g/kg/day for healthy adults.⁴⁵ The safe level of protein intake was identified as the 97.5th percentile of the population distribution of the suggested requirement (i.e., 0.83 g/kg/day). However, these estimates are for completely digested and high-quality protein. Based on calculations of the protein quality and digestibility of proteins in an Indian mixed vegetarian diet (with milk products), the protein digestibility corrected amino acid score (PDCAAS) of these mixed proteins comes to about 85%. Hence, the adequate protein intake would be about 1 g/kg/day (requirement divided by the

TABLE 7. RECOMMENDED OIL COMBINATIONS IN INDIAN DIETS (OILS IN 1:1 PROPORTION)

Oil containing LA + oil containing both LA and ALA	Oil containing high LA + oil containing moderate or low LA
Groundnut/sesame/rice bran/cottonseed + mustard	Sunflower/safflower + palmolein/olive
Groundnut/sesame/rice bran/cottonseed + canola	Safflower/sunflower + groundnut/sesame/rice bran
Groundnut/sesame/rice bran/cottonseed + soybean	
Palmolein + soybean	
Safflower/sunflower + palmolein + mustard	

Data are from Ghafoorunissa.⁴² Health-promoting non-glyceride components include all oils, vitamin E, and plant sterols; sesame oil includes lignans; rice bran oil includes tocotrienols and oryzanols; and palmolein includes tocotrienols. Oils to be used for frying include palmolein/palm oil, groundnut, rice bran, sesame, and cottonseed as single/blends (home/commercial).

ALA, α -linolenic acid; LA, linoleic acid.

TABLE 8. QUANTITIES OF FOODS REQUIRED FOR FURNISHING 0.1 G OF α -LINOLENIC ACID

Foods	Grams
Cereal/millet	
Wheat and pearl millet (<i>bajra</i>)	70
Pulses	
Black gram (<i>kala chana</i>), kidney beans (<i>rajmah</i>), and cowpea (<i>lobia</i>)	20
Vegetables	
Green leafy	60
Purslane (<i>lunia</i>)	25
Other vegetables	400
Fruits*	
Raspberry	80
Avocado	90
Guava	100
Strawberry	155
Kiwi	240
Spices	
Fenugreek seeds (<i>methi</i>)	5
Mustard (<i>sarson</i>)	1
Unconventional oil seeds	
Flaxseed (<i>alsi</i>)	0.5
Perilla seeds (Bhanjira)	0.3

Data are from Ghafoorunissa⁴³ unless otherwise indicated.

*U.S. Department of Agriculture Nutrient Data Base.⁴⁴

PDCAAS score). It is also relevant to consider the relationship of the protein energy with the total dietary energy (protein:energy ratio). Protein requirement usually does not change (unless body weight changes); however, the energy requirement can change, thus changing protein:energy ratio. Simply adding protein into the daily diet will not improve muscle mass. Physical exercise is required to improve muscle mass; because additional exercise will increase the energy requirement, the protein:energy ratio is not likely to change significantly.

Key principles

1. Optimal protein intake is required for healthy growth and prevention of protein malnutrition.
2. Utilization of protein only occurs with a diet adequate in micronutrients.
3. Usually, there is no need to recommend diets with a protein:energy ratio of >15% for growth, or even when enhancement or preservation of skeletal muscle mass is required.

Recommendations

1. Protein intake should be based on body weight. This should be 1 g/kg/day, considering the quality of protein in a usual Indian vegetarian diet.
2. In conjunction with energy intake, the protein intake should provide 10–15% of the total calories/day in sedentary to moderately active individuals.
3. Recommended protein sources:
 - a. Non-vegetarian: egg white, fish, and lean chicken.
 - b. Vegetarian: soya, pulses, whole grams (*channa*, *rajma*, *green gram*, etc.), milk, and low fat dairy products.

Salt

Dietary sodium content is an important determinant of individual and population levels of blood pressure. Reducing dietary sodium consumption reduces blood pressure and vascular risk.⁴⁶ The prevalence of hypertension is increasing in urban India.¹⁵ Some studies suggest an increasing trend of high salt intake (8.5–9 g/day), which is considerably higher than that recommended by the World Health Organization (5 g/day).⁴⁷ Excess salt intake in Asian Indians may be due to intake of Indian *pickles* (fruits and vegetables pickled and preserved in salt and oil), *papad* (indigenous savory salty snack), *namkeens* (salty fried snacks), and *chutneys* (condiments, usually involving a fresh, chopped primary vegetable or fruit with added seasonings mixed with salt used to enhance taste). Furthermore, consumption of salted potato chips by children may be an additional important contributor to high salt consumption.⁴⁸

Key principle

Consumption of salt should be restricted in accordance to currently prevailing international guidelines.

Recommendations

1. Salt intake should be less than 5 g of sodium chloride (or about 2 g of sodium)/day.³¹
2. Addition of extra salt at the dining table should be avoided.
3. Dietary intake of sodium from all sources (*pickles*, *chutneys*, *namkeens*, *papads*, bakery items, potato chips, popcorn, salty biscuits, preserved meat products, other pre-prepared and preserved foods, soups, cheese, and fast foods) should be limited. Avoid processed foods that have high salt content.
4. Reading of food labels to determine sodium content of the commercial foods should be encouraged. Sodium in such foods may be added in such foods in the form of sodium benzoate, monosodium glutamate, baking powder, and baking soda.

Sugar and Artificial Sweeteners

Simple sugars promote a positive energy balance. Total energy increases when the energy density of the diet is increased by sugars or fat.^{49,50} Although common traditional beverages consumed in Asian Indian households include lemon water (with sugar and salt), tea (with sugar), and *lassi* (a beverage made by blending yogurt with water and salt/sugar), recently an increasing consumption of sweetened carbonated beverages has been seen especially by adolescents. On an average, about 1.8 cans of cola per week (540 mL/week) per person consumption has been reported in urban adolescents (1 can [300 mL] = 132 kcal and 33–40 g of sugar).⁵¹

Key principle

Intake of simple sugars should be restricted.

Recommendations

1. Free sugars should be less than 10% of total calories/day, which includes all added sugars and sugars present in honey, syrups, and fruit juices.³¹

2. Alternatives to sweetened beverages can be water, skimmed buttermilk, tender coconut water, and low fat milk.
3. Indian sweets (*halwa* [a gelatinous sweet dish made from grain flour, *ghee*, sugar, and nuts] and *kheer* [a sweet dish made from boiling rice with milk, sugar, cardamoms, saffron, and nuts], puddings, ice creams, sweetened biscuits, cakes, pastries, and baked goods are high in added sugars and should be restricted.
4. Encourage reading of food labels to determine sugar content. Some of the names in the ingredients list for the presence of added sugars include brown sugar, corn syrup, dextrose, honey, malt syrup, sugar, molasses, and sucrose.

Artificial sweeteners could be used in moderation. However, these do not contain any beneficial nutrients, and the long-term health benefit, if any, is not clear in individuals without diabetes. The Food and Drug Administration has approved five artificial sweeteners: saccharin (e.g., Sweetex [Reckitt Benckiser, Slough, UK]), aspartame (e.g., Equal [Cadila Healthcare Ltd., Ahmadabad, India]), acesulfame-K (e.g., Sweet One [Hugestone Enterprise Co., Ltd., Jiangsu, China]), neotame (e.g., NutraSweet [NutraSweet Co., Chicago, IL]) (both acesulfame-K and neotame are used in beverages, dairy products, pharmaceutical products, chewing gum, etc.), and sucralose (Sugar Free Natura [Acme Remedies, Malvern, PA]) as safe.⁵²⁻⁵⁴ Although doubts have been raised regarding safety of saccharin, the Food and Drug Administration has approved it to be used in limited quantity because of low price, good shelf life, and heat stability. Stevia (e.g., Stevia [Rigil Blotech (P) Ltd., New Delhi, India]) and some sugar alcohols (e.g., erythritol [Zsweet®, Zsweet, Dublin, UK], etc.) have been approved by the Food and Drug Administration under Generally Recognized as Safe status.

Other Dietary Habits

Water

Water is necessary for metabolism and for physiological functions in the body and is also a source of essential minerals, including calcium, magnesium, and fluoride. Fluid requirements vary depending on individuals and specific population.⁵⁵ Advocacy by some individuals on drinking excess quantities of water is scientifically untenable. However, increased water intake is recommended under special circumstances, such as vigorous work and outdoor activity in hot climates.⁵⁶

Recommendation. An individual should have 1.5–2 L (8–10 glasses) of water every day; intake could be increased in hot climates.

Alcohol

According to the guidelines of the 2001 National Cholesterol Education Program, Adult Treatment Panel III⁵⁷ and in 2006 the American Heart Association,⁵⁸ alcohol intake should be limited to one drink per day (equivalent to 30 mL of whisky/gin/vodka, 120 mL of wine, or 300 mL of beer) for women and two such drinks per day for men. However, alcohol should not be taken if serum triglycerides are 500 mg/dL or above⁵⁹ and in the presence of significant liver dysfunction. Most of these studies have evaluated white Caucasian subjects, and whether these results can be extrapolated in Asian Indians, who already have a high prevalence of fatty liver, is not clear.

Recommendations. Regular excessive intake of alcohol is harmful. Until more data are available for Asian Indians, nonconsumers of alcohol should not have alcohol; however, individuals taking a small quantity of alcohol should not be discouraged.

Food choices while eating out

The report of the Joint World Health Organization/Food and Agriculture Organization Expert Consultation on “Diet, Nutrition, and the Prevention of Chronic Diseases” clearly stated that eating behaviors linked to overweight/obesity and consequent chronic diseases include snacking/eating frequency, binge-eating patterns, and eating out.³¹

Recommendations.

1. Choose healthy snacks.
2. Follow the healthy dietary guidelines while eating out as described above.
3. Smaller-sized portion should be preferred.
4. Avoid sweetened carbonated drinks and commercially available high calorie drinks and opt for beverages like buttermilk, coconut water, fresh lime with water, etc.

Meal portion and times

Psychological parameters of eating patterns also seem to influence risk of obesity, with the “flexible restraint” pattern having lower risk of overweight than the “rigid restraint/periodic disinhibition” pattern. In the same context, a high frequency of eating shows a negative relationship with weight gain.³¹

Recommendations.

1. Small frequent meals at regular intervals should be taken.
2. The gap between two meals should be 3–4 hours.

Regular breakfast

Eating breakfast plays a significant role in effective weight control. The National Health and Nutrition Examination Survey III (1988–1994) data showed that people who skip breakfast had higher mean body mass index than those who did not.⁶⁰ Breakfast skipping leads to excessive/imbalanced eating later in day, a dietary pattern associated with obesity.

Recommendation. A healthy regular breakfast should be an essential part of the meal plan.

Cooking methods

Correct cooking methods can minimize intake of fats in the diet.

Recommendations. To minimize dietary fat intake, deep-frying of foods should be avoided.⁶¹ If such a cooking method is unavoidable, then methods (such as using a blotting paper) should be used to drain out the excess oil from the fried cooked food.

Instead, roasting or grilling not only eliminates added oil but may also allow any fat already present in food to drip away. Hence, the following methods are recommended:

boiling, steaming, roasting, grilling, stewing, broiling, or making sautéed and poached preparations.

Conclusions

In comparison with the previous guidelines of the National Institute of Nutrition, the Consensus Group recommends a reduction in the intake of carbohydrate (50–60% of total energy/day), preferential intake of complex carbohydrates and low GI foods, higher intake of fiber (25–40 g/day), lower intake of saturated fats (less than 10% of total energy/day), optimal ratio of essential fatty acids/day (LA 5–8% and ALA 1–2 % of total energy, optimal ratio of LA/ALA 5–10, *cis* MUFAs 10–15%, TFAs <1% of total energy), slightly higher protein intake (10–15% of total energy/day), lower intake of salt (5 g/day), and restricted intake of sugar (less than 10% of total energy/day). Although these guidelines are applicable to Asian Indians in any geographical setting, they are particularly applicable to those residing in urban and in semi-urban areas. Proper application of these guidelines will help curb the rising “epidemics” of the metabolic syndrome, T2DM, and CVD in Asian Indians.

Appendix 1

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